Time: 1 hr. Paper Code: CEN-504	B. Tech. (V Semester) 1st Sessional Test 2024-25 Object Oriented Programming	Max. Marks: 15

Note: Attempt All Questions

Q1. **CO1**	Design a program in C++ that models the interaction between students and courses in a university system. The system should allow students to enroll in courses and withdraw from them, while each course has a maximum capacity of students that can be enrolled. The system should use two classes, *Student* and *Course*, which communicate with each other. The course will keep track of enrolled students and allow the addition and removal of students based on their IDs.	[5]

Q2.
CO2
	Implement a class *Time* that contains three integer data members to represent hours, minutes, and seconds. Write the necessary code such that the following driver function runs correctly and performs the desired operations.
	`int main() {`
	`  Time T1(2, 45, 50);`
	`  Time T2(1, 30, 40);`
	`  Time T3 = T1 + T2;`
	`  cout << "T1 = " << T1 << endl;`
	`  cout << "T2 = " << T2 << endl;`
	`  cout << "T3 = " << T3 << endl;`
	`  return 0;`
	`}`
		[5]

Q3. **CO3**	Differentiate between compile time and run time polymorphism. Demonstrate their implementation with the help of relevant code in C++.	[5]